

2

NUMBER AND ALGEBRA

GRAPHING LINES

Straight lines are an important part of our environment. We play sport on courts using parallel and perpendicular lines, and skyscrapers would not be standing without straight lines. We can also use straight lines to model different types of data and predict future outcomes.



Chapter outline

	Working mathematically				
2.01 Length, midpoint and gradient of an interval	U	F	PS	R	C
2.02 Parallel and perpendicular lines	U	F			
2.03 Graphing linear equations	U	F		R	C
2.04 The gradient–intercept equation $y = mx + c$	U	F		R	
2.05 The general equation $ax + by + c = 0$*	U	F		R	C
2.06 Finding the equation of a line	U	F		R	C
2.07 Applying linear equations	U	F		R	C
2.08 Equations of parallel and perpendicular lines	U	F		R	C

*PATH

Wordbank

general form Any linear equation expressed as $ax + by + c = 0$, where a, b and c are integers and a is positive

gradient The steepness of a line or interval, measured by the fraction $\frac{\text{rise}}{\text{run}}$

gradient–intercept form Any linear equation expressed as $y = mx + c$, where m is the gradient and c is the y-intercept

linear equation An equation whose graph is a straight line

parallel lines Lines that point in the same direction and have the same gradient

perpendicular lines Lines that cross at right angles (90°) and have gradients whose product is −1

x-intercept The x-value at which a graph cuts the x-axis

y-intercept (or **vertical intercept**) The y-value at which a graph cuts the y-axis

Quiz
Wordbank 2

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communication

Videos (15):

2.01 The gradient of a line • Coordinate geometry • Distance, midpoint and gradient formulas

2.02 Gradients of perpendicular lines

2.03 Graphing linear equations • Testing if a point lies on a line • Horizontal and vertical lines

2.04 Gradient and y-intercept of a line • The gradient-intercept formula • Graphing $y = mx + c$

2.05 The general equation $ax + by + c = 0$

2.06 Finding the equation of a line

2.07 Applying linear functions

2.08 Equations of parallel lines • Equations of perpendicular lines

PhET interactives (4):

2.03 Graphing lines • Function builder: Basics • Function builder

2.04 Graphing slope-intercept

Quizzes (5):

- Wordbank 2
- SkillCheck 2
- Mental skills 2
- Language of maths 2
- Test yourself 2

Skillsheets (2):

SkillCheck Pythagoras' theorem

2.03 Graphing linear equations

Worksheets (11):

2.01 Gradient, midpoint and distance • Intervals match-up • Gradient between 2 points

2.03 A page of lines • A page of number planes • Graphing linear equations • Straight-line equations

2.05 Parallel and perpendicular lines

2.06 Finding the equation of a line • A page of lines

2.08 Writing equations of lines

Mind map Graphing lines

Puzzles (9):

2.01 Finding coordinates for given segment lengths

2.02 Gradients of parallel and perpendicular lines

2.04 Equation of a line • Equations in gradient form

2.05 Linear equations code puzzle

2.06 Straight-line equations

2.08 Linear equations match-up • Equations of parallel lines

Language of maths Graphing lines crossword

Technology (1):

2.01 Midpoint and distance between 2 points

Spreadsheets (2):

2.01 Midpoint and distance between 2 points

2.04 Drawing straight lines: $y = mx + c$

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

In this chapter you will:

- find the distance between 2 points on a number plane
- find the midpoint and gradient of an interval on a number plane
- find the properties of the gradients of parallel and perpendicular lines
- (PATH) find the x- and y-intercepts of a line from its equation
- graph a linear equation on a number plane
- test whether a point lies on a line
- use the gradient–intercept equation of a straight line $y = mx + c$
- (PATH) convert between the general form of a linear equation $ax + by + c = 0$ and the gradient–intercept equation
- find the equation of a line from its graph
- solve problems that can be modelled by linear equations
- find the equation of a line that is parallel or perpendicular to a given line

SkillCheck ANSWERS ON P. 577

1 For this number plane, find:

a the midpoint of interval BC

b the midpoint of interval HE

c the length of interval GC

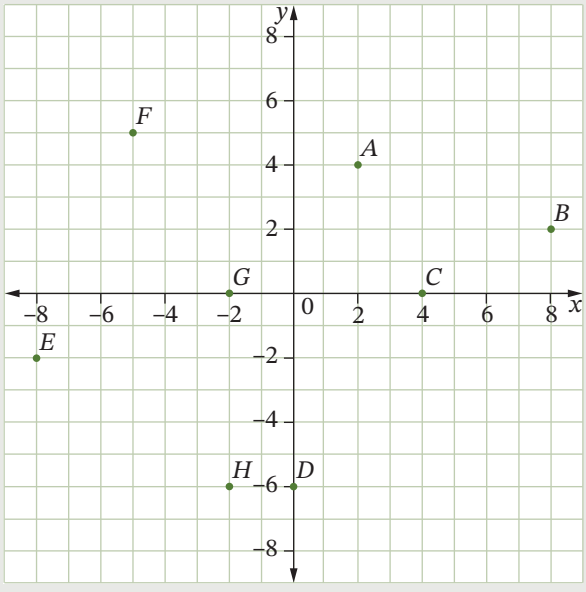
d the length of interval GH

e the lengths of AC and BC, correct to one decimal place

f the type of triangle $\triangle ABC$ is

g the gradient of GE

h the gradient of EH



2 For each linear equation, copy and complete the table of values and graph the equation.

a $y = x - 3$

x	0	1	2	3
y				

b $y = 3x + 2$

x	−2	−1	0	1
y				

c $y = 1 - 2x$

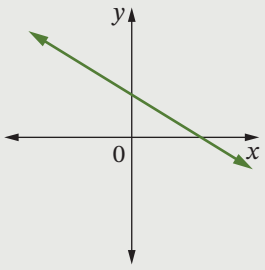
x	−1	0	1	2
y				

Quiz SkillCheck 2

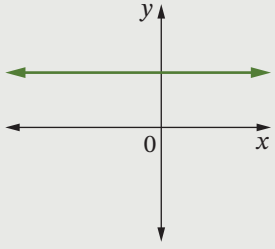
Skillsheet Pythagoras' theorem

3 State whether each line's gradient is positive, negative or neither.

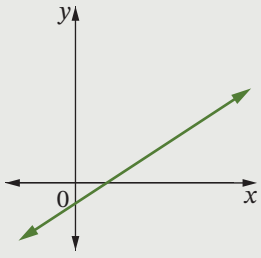
a



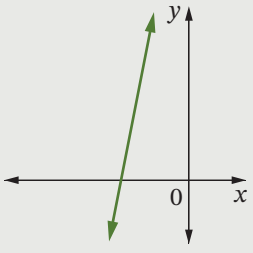
b



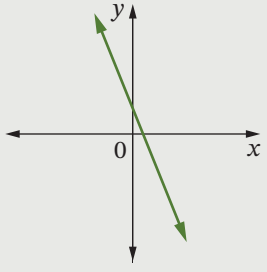
c



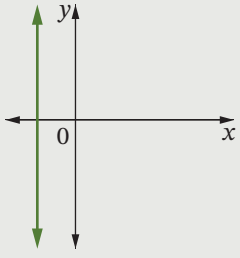
d



e



f



2.01 Length, midpoint and gradient of an interval

Worksheets
Gradient, midpoint and distance

Intervals match-up

Gradient between 2 points

Technology
Midpoint and distance between 2 points

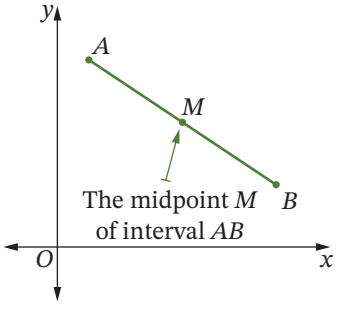
Spreadsheet
Midpoint and distance between 2 points

Video
The gradient of a line

The **length** of an interval AB (or the **distance** between A and B) can be calculated using Pythagoras' theorem if we know the coordinates of A and B.

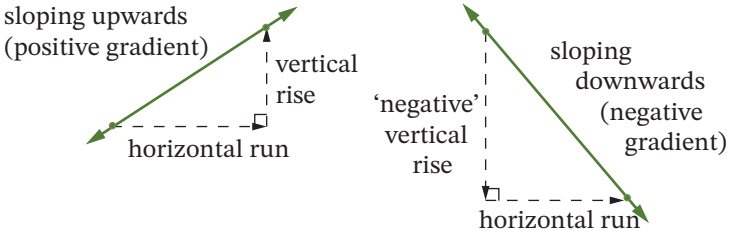
The **midpoint** of an interval AB is the point in the middle of AB or halfway between A and B.

- Its x-coordinate is the average of the x-coordinates of A and B.
- Its y-coordinate is the average of the y-coordinates of A and B.



The **gradient** of an interval measures its steepness. It is given by the formula:

$$m = \frac{\text{vertical rise}}{\text{horizontal run}} = \frac{\text{rise}}{\text{run}}$$



- A line **sloping upwards** has a **positive rise** and **a positive gradient**.
- A line **sloping downwards** has a **negative rise** and a **negative gradient**.
- The **run** is always **positive**.

9780170479622

Example 1

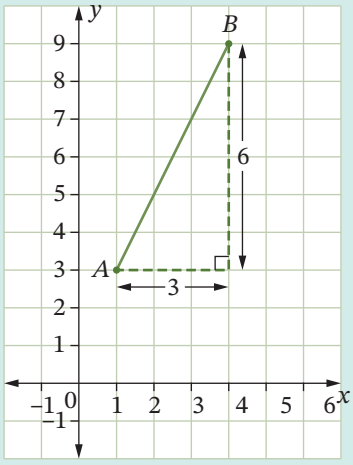
For the interval joining each pair of points given, find:

- **i** the length of the interval, correct to one decimal place
- **ii** the midpoint of the interval
- **iii** the gradient of the interval.

a $A(1, 3)$ and $B(4, 9)$ **b** $P(-5, 8)$ and $Q(3, 6)$

Solution

a **i** Draw a right-angled triangle on the number plane, with AB as the hypotenuse.



$\text{Height} = 9 - 3 = 6$ Difference between y-coordinates

$\text{Base} = 4 - 1 = 3$ Difference between x-coordinates

$AB^2 = 6^2 + 3^2$ by Pythagoras' theorem

$= 45$

$AB = \sqrt{45}$

$= 6.7082...$

≈ 6.7 units

ii For $A(1, 3)$ and $B(4, 9)$, the average of the x-coordinates is $\frac{1+4}{2} = 2\frac{1}{2}$.

The average of the y-coordinates is $\frac{3+9}{2} = 6$.

$\therefore$ The midpoint of AB is $\left(2\frac{1}{2}, 6\right)$.

From the diagram above, a midpoint at $\left(2\frac{1}{2}, 6\right)$ looks reasonable.

iii The rise is 6 units.

The run is 3 units.

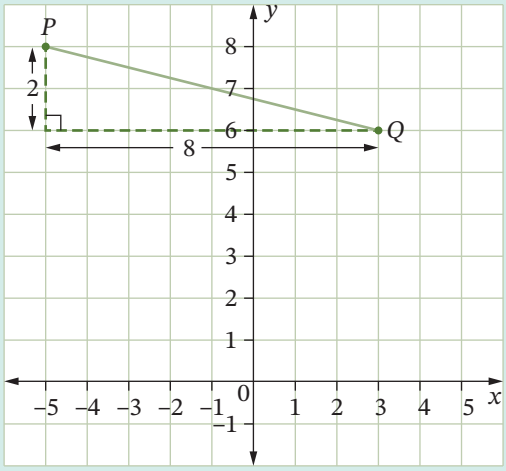
$$m = \frac{\text{rise}}{\text{run}}$$
$$= \frac{6}{3}$$
$$= 2$$

The gradient is positive, the line is sloping upwards.

b i Draw a right-angled triangle on the number plane with PQ as the hypotenuse.

The height of the triangle is 2 units.

The base of the triangle is 8 units.



$PQ^2 = 2^2 + 8^2$ by Pythagoras' theorem

$= 68$

$PQ = \sqrt{68}$

$= 8.2462...$

≈ 8.2 units

ii For $P(-5, 8)$ and $Q(3, 6)$, the average of the x-coordinates is $\frac{-5+3}{2} = -1$.

From the diagram above, a midpoint at $(-1, 7)$ looks reasonable.

The average of the y-coordinates is $\frac{8+6}{2} = 7$.

$\therefore$ The midpoint of PQ is $(-1, 7)$.

iii The rise is -2 units.

Line slopes downwards

The run is 8 units.

$$m = \frac{\text{rise}}{\text{run}}$$
$$= \frac{-2}{8}$$
$$= -\frac{1}{4}$$

The gradient is negative, the line slopes downward.

Optional: The distance, midpoint and gradient formulas

PATH

Video
Coordinate geometry

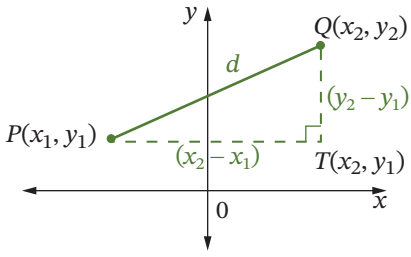
2.01

The methods for finding the length, midpoint and gradient of an interval can each be summarised by a formula.

The **distance formula** is used to calculate the distance (d) between 2 points $P(x_1, y_1)$ and $Q(x_2, y_2)$, that is, the length of the interval PQ.
By Pythagoras' theorem:

$d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$

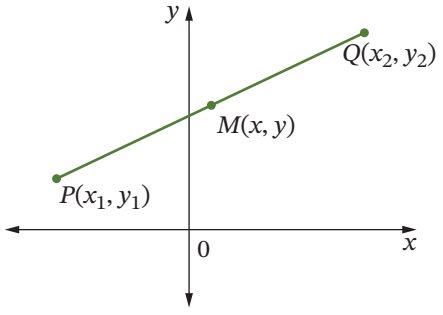
$\therefore d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$



The **midpoint formula** gives the coordinates of the point M, the midpoint of the interval joining $P(x_1, y_1)$ and $Q(x_2, y_2)$:

$M(x, y) \equiv \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$

We've used '≡' ('is identical to') rather than '=' because we are referring to the point (x, y), not a number.



The **gradient formula** gives the gradient of the interval or line joining $P(x_1, y_1)$ and $Q(x_2, y_2)$.

Vertical rise = difference in y-coordinates $= y_2 - y_1$

Horizontal run = difference in x-coordinates $= x_2 - x_1$

Gradient, $m = \frac{\text{difference in } y}{\text{difference in } x} = \frac{y_2 - y_1}{x_2 - x_1}$

Example 2

For the interval joining $P(-5, 8)$ and $Q(3, 6)$ from Example **1b**, use a formula to find:

a the length of the interval, correct to one decimal place

b the midpoint of the interval

c the gradient of the interval.

Solution

For $P(-5, 8)$ and $Q(3, 6)$: $x_1 = -5, y_1 = 8, x_2 = 3, y_2 = 6$.
(x_1, y_1) (x_2, y_2)

a $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ Apply the distance formula

$= \sqrt{(3 - (-5))^2 + (6 - 8)^2}$

$= \sqrt{68}$

$= 8.2462\ldots$

≈ 8.2 units

Video
Distance, midpoint and gradient formulas

Puzzle
Finding coordinates for given segment lengths

PATH

b $M(x, y) \equiv \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$ — Apply the midpoint formula

$\equiv \left(\frac{-5+3}{2}, \frac{8+6}{2}\right)$

$\equiv (-1, 7)$

c $m = \frac{\text{difference in } y}{\text{difference in } x}$ — Apply the gradient formula

$= \frac{y_2 - y_1}{x_2 - x_1}$

$= \frac{6-8}{3-(-5)}$

$= \frac{-2}{8}$

$= -\frac{1}{4}$

Example 3

a Plot the points $A(0, 6)$, $B(5, 6)$, $C(5, 2)$ and $D(-4, 2)$ on a number plane and join them to make the quadrilateral $ABCD$.

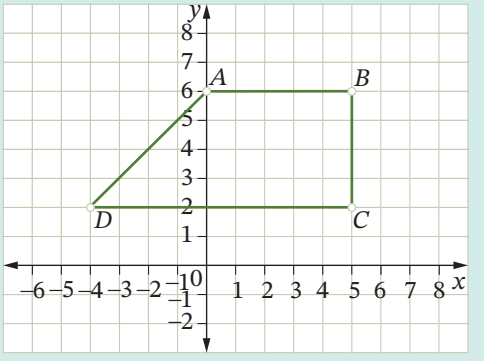
b What type of quadrilateral is $ABCD$?

c Find the exact length of AD.

d Hence find the perimeter of $ABCD$, correct to 2 decimal places.

Solution

a Join the points in the correct order



b Since $AB \parallel CD$, the quadrilateral is a trapezium.

c $AD^2 = 4^2 + 4^2$

$= 32$

$AD = \sqrt{32}$ units — In exact surd form

d By counting grid squares, $AB = 5$, $BC = 4$, $CD = 9$.

Perimeter of $ABCD = 5 + 4 + 9 + \sqrt{32}$

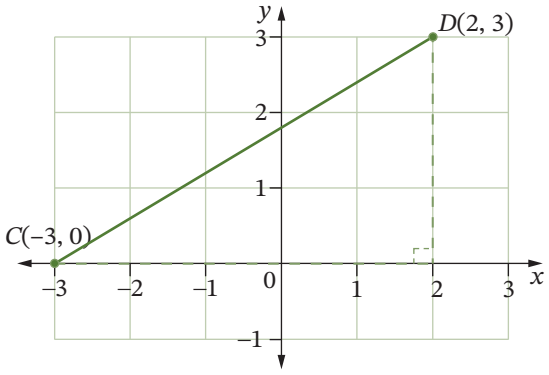
$= 23.656...$

≈ 23.66 units

EXERCISE 2.01 ANSWERS ON P. 578

Length, midpoint and gradient of an interval U F PS R C

Questions **1**, **2** and **3** refer to this diagram of interval *CD*.



EXAMPLES 1, 2

1 What is the length of interval *CD*? Select the correct answer **A**, **B**, **C** or **D**.

A 2 units **B** 5.8 units **C** 3.2 units **D** 8 units

2 What is the midpoint of *CD*? Select **A**, **B**, **C** or **D**.

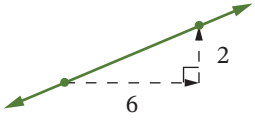
A $(-1, 3)$ **B** $(-5, 3)$ **C** $(-0.5, 1.5)$ **D** $(-2.5, 1.5)$

3 What is the gradient of *CD*? Select **A**, **B**, **C** or **D**.

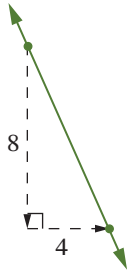
A $\frac{3}{5}$ **B** $-\frac{3}{5}$ **C** $-\frac{5}{3}$ **D** $\frac{5}{3}$

4 Calculate the gradient of each line.

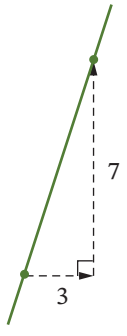
a



b



c



5 For the interval joining each pair of points given, find:

i the length of the interval, correct to one decimal place

ii the midpoint of the interval

iii the gradient of the interval.

a $A(5, 3)$ and $B(7, 2)$ **b** $J(-1, 0)$ and $K(8, 6)$ **c** $M(0, -3)$ and $N(-5, 2)$

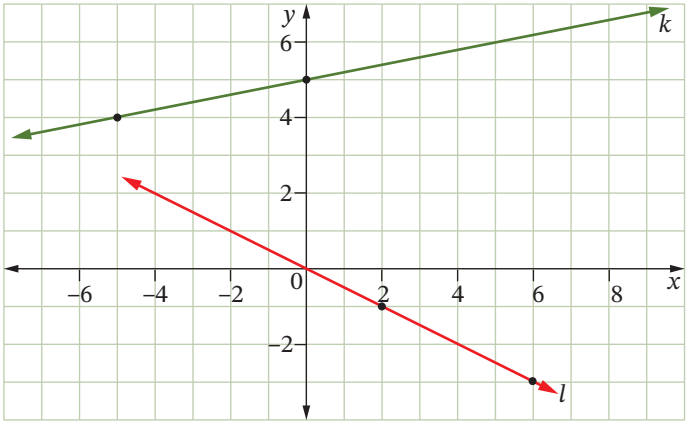
d $R(-3, -6)$ and $S(4, -9)$ **e** $A(-7, 2)$ and $B(-5, -8)$ **f** $U(3, -2)$ and $V(7, 2)$

Foundation Standard Complex

6 Calculate, in exact (surd) form, the distance between each pair of points.

a $(-8, -1)$ and $(0, 4)$ **b** $(12, -6)$ and $(-1, -1)$ **c** $(7, -2)$ and $(-2, -3)$

7 Find the gradient of the lines labelled k and l.



8 Which expression gives the y-coordinate of the midpoint of the interval joining points $(3, 8)$ and $(-1, 5)$? Select **A**, **B**, **C** or **D**.

A $\frac{-1+5}{2}$ **B** $\frac{8+5}{2}$ **C** $\frac{8-5}{2}$ **D** $\frac{5-8}{2}$

EXAMPLE 3

9 The vertices of triangle ABC are $A(-1, -1)$, $B(1, 3)$ and $C(3, 1)$. PS R C

a Draw $\triangle ABC$ on a number plane.

b Find the exact length of each side of the triangle.

c Are any sides of the triangle equal in length?

d What type of triangle is ABC?

e Find the perimeter of $\triangle ABC$, correct to one decimal place.

10 The vertices of quadrilateral $KLMP$ are $K(1, 6)$, $L(7, 2)$, $M(3, -4)$ and $P(-3, 0)$. PS R C

a Draw the quadrilateral on a number plane.

b What type of quadrilateral is $KLMP$?

c Find the gradients of sides KL and PM.

d Find the gradients of sides KP and LM.

e What do you notice about the gradients of the opposite sides of this quadrilateral? What does that mean about those sides?

f Find the exact length of each side of $KLMP$.

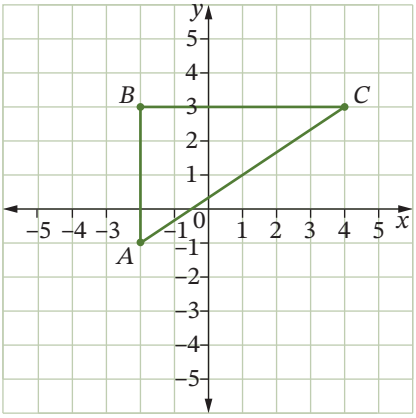
g Find the perimeter of $KLMP$, correct to one decimal place.

h Find the area of $KLMP$.

Foundation Standard Complex

9780170479622

11 This diagram shows a right-angled triangle with vertices $A(-2, -1)$, $B(-2, 3)$ and $C(4, 3)$. R C



a Copy the diagram and find the coordinates of P and Q, the midpoints of BA and BC respectively. Mark P and Q on your diagram.

b Calculate, correct to one decimal place, the lengths of PQ and AC. What do you notice about your answers?

c Find the gradients of PQ and AC. What do you notice about your answers?

Investigation

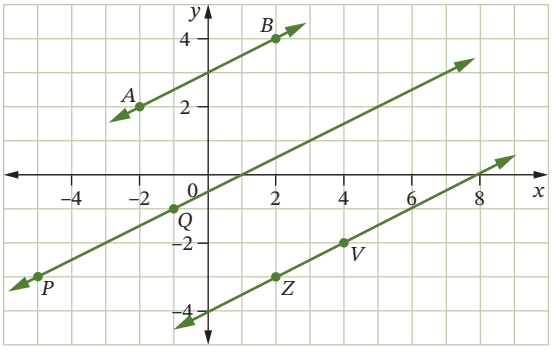
Parallel and perpendicular lines

1 These 3 lines are parallel. Calculate the gradient of:

a AB

b PQ

c ZV

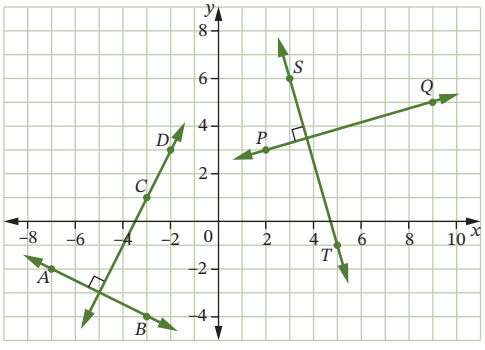


2 What can you conclude about the gradients of parallel lines?

3 This diagram shows 2 pairs of perpendicular lines. $AB \perp CD$ and $PQ \perp ST$. Calculate the gradient of:

a AB **b** CD

c PQ **d** ST



4 Is there a relationship between:

a the gradients of AB and CD? **b** the gradients of PQ and ST?

5 Calculate the product of:

a the gradients of AB and CD **b** the gradients of PQ and ST

6 What can you conclude about the gradients of perpendicular lines?

Foundation Standard Complex

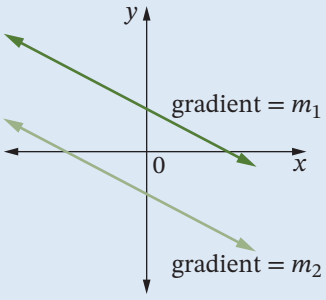
2.02 Parallel and perpendicular lines

Puzzle
Gradients of parallel and perpendicular lines

Parallel lines

Parallel lines have the same gradient.

If 2 lines with gradients m_1 and m_2 are parallel, then $m_1 = m_2$.





Rost9/Shutterstock.com

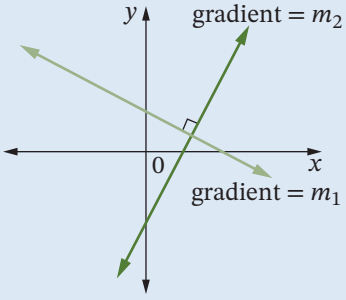
Perpendicular lines

Perpendicular lines have gradients with a product of -1.

If 2 lines with gradients m_1 and m_2 are perpendicular, then

$$m_1 \times m_2 = -1 \text{ or } m_2 = -\frac{1}{m_1}.$$

Note that m_2 is the **negative reciprocal** of m_1.



Example 4

State whether each pair of gradients represent parallel lines, perpendicular lines or neither.

a $m_1 = \frac{1}{2}, m_2 = 2$ **b** $m_1 = 0.4, m_2 = \frac{2}{5}$ **c** $m_1 = 1\frac{3}{5}, m_2 = -\frac{5}{8}$

Solution

a $m_1 \neq m_2$ so the lines are not parallel.

$$m_1 \times m_2 = \frac{1}{2} \times 2$$
$$= 1$$
$$\neq -1$$

so the lines are not perpendicular.

$\therefore$ The lines are neither parallel nor perpendicular.

b $m_2 = \frac{2}{5} = 0.4$

$m_1 = m_2$

$\therefore$ The lines are parallel.

c $m_1 = 1\frac{3}{5} = \frac{8}{5}$

$$m_1 \times m_2 = \frac{8}{5} \times \left(-\frac{5}{8}\right)$$
$$= -1$$

$\therefore$ The lines are perpendicular.

Example 5

Find the gradient of a line that is perpendicular to a line with gradient:

a 2 **b** -3 **c** $\frac{3}{4}$ **d** -0.6

Solution

a $m_1 = 2$

$m_2 = \frac{-1}{m_1}$ for perpendicular lines

$= \frac{-1}{2}$ The negative reciprocal of m_1.

$= -\frac{1}{2}$

The gradient is $-\frac{1}{2}$.

b $m_1 = -3$

$m_2 = \frac{-1}{m_1}$

$= \frac{-1}{-3}$

$= \frac{1}{3}$

The gradient is $\frac{1}{3}$.

c $m_1 = \frac{3}{4}$

$m_2 = \frac{-1}{m_1}$

$= \frac{-1}{\left(\frac{3}{4}\right)}$

$= -\frac{4}{3}$

The gradient is $-\frac{4}{3}$.

d $m_1 = -0.6 = -\frac{3}{5}$

$m_2 = \frac{-1}{\left(-\frac{3}{5}\right)} = \frac{5}{3}$

The gradient is $\frac{5}{3}$.

2.02

Video
Gradients of perpendicular lines

Example 6

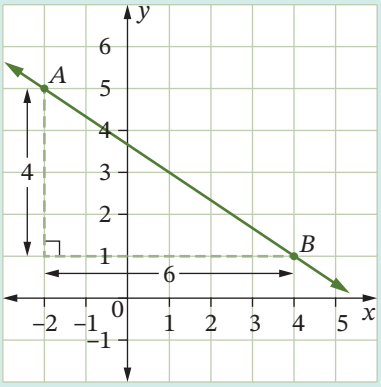
A line passes through the points $A(-2, 5)$ and $B(4, 1)$. What is the gradient of a line:

a parallel to AB? **b** perpendicular to AB?

Solution

Find the gradient of AB by calculating the rise and run.

Sometimes drawing a diagram helps you to understand the problem.



Rise $= 1 - 5 = -4$ Difference between y-coordinates

Run $= 4 - (-2) = 6$ Difference between x-coordinates

Gradient $AB = \frac{-4}{6} = -\frac{2}{3}$ $\frac{\text{rise}}{\text{run}}$

a Any line parallel to AB will have the same gradient as AB.

$\therefore m = -\frac{2}{3}$

b The gradient of a line perpendicular to AB will be given by:

$$m = \frac{-1}{\left(-\frac{2}{3}\right)} = \frac{3}{2}$$

EXERCISE 2.02 ANSWERS ON P. 578

Parallel and perpendicular lines U F

EXAMPLE 4

1 State whether each pair of gradients represent parallel lines, perpendicular lines or neither.

a $m_1 = \frac{1}{4}, m_2 = 4$ **b** $m_1 = 3, m_2 = -\frac{1}{3}$ **c** $m_1 = 0.5, m_2 = \frac{1}{2}$

d $m_1 = \frac{2}{7}, m_2 = \frac{7}{2}$ **e** $m_1 = \frac{3}{10}, m_2 = 0.3$ **f** $m_1 = 1\frac{1}{5}, m_2 = -\frac{6}{5}$

2 Find the gradient of a line that is parallel to a line with gradient:

a 4 **b** -2 **c** $\frac{1}{3}$ **d** -0.2

Foundation Standard Complex

9780170479622

EXAMPLE 5

3 Find the gradient of a line that is perpendicular to a line with gradient:

a 1 **b** -6 **c** -1.5 **d** $\frac{5}{2}$

4 What is the gradient of a line that is perpendicular to a line with a gradient of 0.8? Select the correct answer **A**, **B**, **C** or **D**.

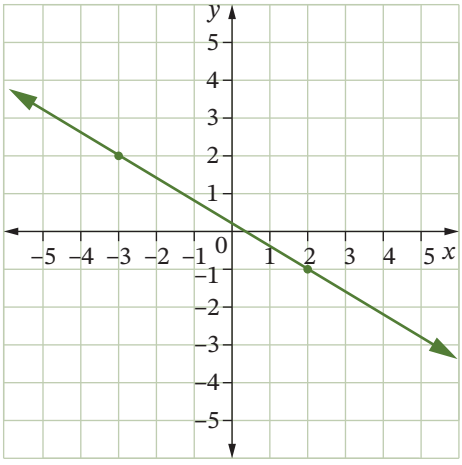
A 0.2 **B** -0.2 **C** 1.25 **D** -1.25

5 What is the gradient of a line that is parallel to a line that goes through $P(0, 3)$ and $Q(5, -2)$? Select **A**, **B**, **C** or **D**.

A 1 **B** -1 **C** $\frac{1}{5}$ **D** $-\frac{1}{5}$

EXAMPLE 6

6 What is the gradient of a line perpendicular to the line shown? Select **A**, **B**, **C** or **D**.



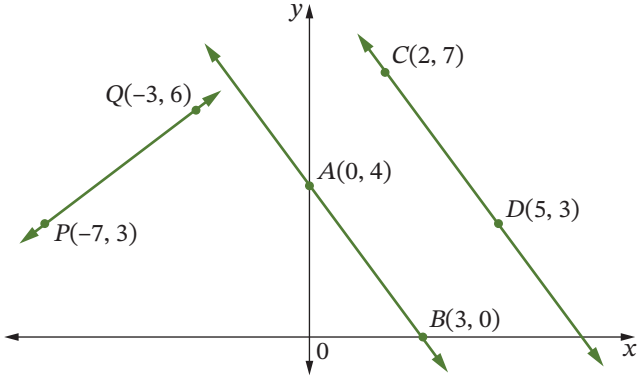
A $\frac{5}{3}$ **B** -5

C $\frac{3}{5}$ **D** $\frac{1}{5}$

7 Calculate the gradient of each line shown and test whether:

a $AB \parallel CD$

b $PQ \perp CD$.



8 A line passes through the points $R(-5, 2)$ and $S(1, 4)$. What is the gradient of a line:

a parallel to RS?

b perpendicular to RS?

Foundation Standard Complex

Technology

Parallel and perpendicular lines

Use dynamic geometry software to find out if sets of linear equations represent parallel or perpendicular lines.

1 Parallel lines

Graph the following lines and use the **Slope/Gradient** function to find their gradients and check whether they are parallel using $m_1 = m_2$.

a $5x - 3y = 0$ and $y = \frac{5x}{3}$

b $x + y + 4 = 0$ and $x + y - 6 = 0$

c $x - 2y = 0$ and $y = 0.5x$

d $y = 5x - 9$ and $5x - y - 1 = 0$

2 Perpendicular Lines

Graph the following lines and use the **Slope/Gradient** function to find their gradients and check whether they are perpendicular using $m_1 \times m_2 = -1$.

a $y = 0.6x + 2$ and $y = \frac{5}{3}x$

b $x - 4y + 1 = 0$ and $y = -4x - 3$

c $3x - 2y = 0$ and $y = -\frac{2x}{3}$

d $y = 2x + 4$ and $x - 2y - 1 = 0$

2.03 Graphing linear equations

Video
Graphing linear equations

Interactives
Graphing lines
Function builder: Basics
Function builder

Worksheets
A page of lines
A page of number planes
Graphing linear equations

Skillsheet
Graphing linear equations

A relationship between 2 variables, x and y, whose graph is a straight line is called a **linear relationship**. The expression of that relationship as an algebraic formula, such as $y = 3x + 2$, is called a **linear equation**.

Example 7

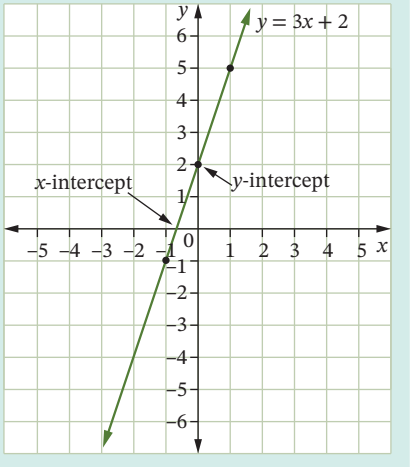
Graph $y = 3x + 2$ on a number plane and find its x- and y-intercepts.

Solution

Complete a table of values. Choose x-values close to 0 for easy calculation and graphing.

x	−1	0	1
y	−1	2	5

Graph (−1, −1), (0, 2) and (1, 5) on a number plane. Rule a straight line through the points, place arrows at each end, and label the line with its equation.



9780170479622

The x-intercept is the x-value where the line cuts the x-axis and can be found by substituting $y = 0$ into the equation $y = 3x + 2$:

$0 = 3x + 2$

$3x = -2$

$x = -\frac{2}{3}$ which agrees with what's shown on the graph

The x-intercept is $-\frac{2}{3}$.

The y-intercept is the y-value where the line cuts the y-axis and can be found by looking at the graph, or looking at the constant term in the equation $y = 3x + 2$, or substituting $x = 0$ into the equation $y = 3x + 2$ (which has already been done for the table of values).

The y-intercept is 2.

In the above example, every point on the line follows the linear equation $y = 3x + 2$.
For example, $(-1, 1)$, $(0, 2)$ and $(1, 5)$ lie on the line and follow the rule $y = 3x + 2$.

There are an infinite number of points that follow the rule. Arrows on both ends of the line indicate that it has infinite length.

Testing if a point lies on a line

A point lies on a line if its (x, y) coordinates satisfy the equation of the line.

Video
Testing if a point lies on a line

Worksheet
Straight-line equations

Example 8

Test whether each point lies on the line $x - 2y = 5$.

a $(17, 6)$ **b** $(8, -4)$

Solution

- Separate the equation into its left-hand side (LHS) and right-hand side (RHS)
- Substitute the coordinates of the point into both sides
- If LHS = RHS, the point satisfies the equation and so lies on the line
- If LHS $\neq$ RHS, the point does not lie on the line.

a Substitute $x = 17, y = 6$ into $x - 2y = 5$.

LHS $= x - 2y$

$= 17 - 2 \times 6$

$= 5$

RHS $= 5$

LHS = RHS, so $(17, 6)$ lies on the line.

b Substitute $x = 8, y = -4$ into $x - 2y = 5$.

LHS $= x - 2y$

$= 8 - 2 \times (-4)$

$= 16$

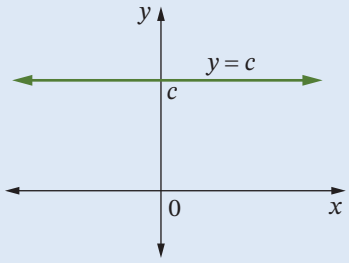
RHS $= 5$

LHS $\neq$ RHS, so $(8, -4)$ does not lie on the line.

Video
Horizontal and vertical lines

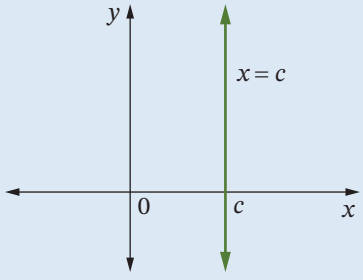
Horizontal and vertical lines

The **equation of a horizontal line** is of the form $y = c$ (where c is a constant number).



Note: The x-axis has equation $y = 0$.

The **equation of a vertical line** is of the form $x = c$ (where c is a constant number).



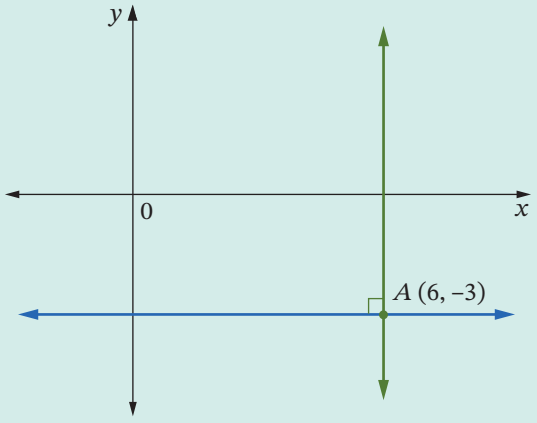
Note: The y-axis has equation $x = 0$.

Example 9

For the graph shown, find the equation of:

a the vertical line

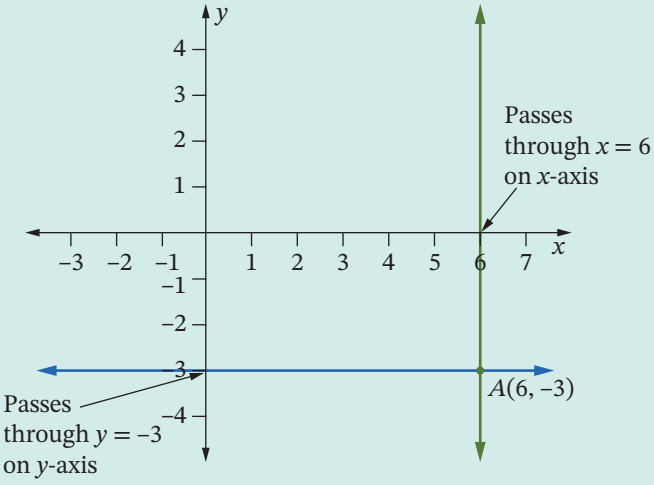
b the horizontal line



Solution

a The vertical line has an x-intercept of 6 and passes though $A(6, -3)$, so its equation is $x = 6$.

b The horizontal line has a y-intercept of -3 and passes though $A(6, -3)$, so its equation is $y = -3$.



9780170479622

Example 10

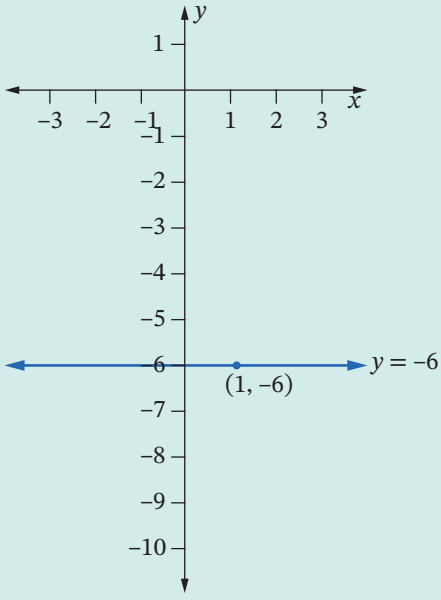
Find the equation of the line that is:

a parallel to the x-axis and passes through the point $(1, -6)$

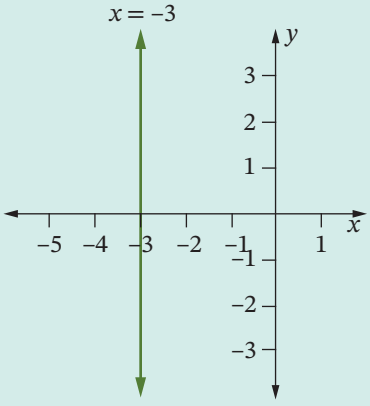
b always 3 units to the left of the y-axis.

Solution

a The equation of the line is $y = -6$.



b The equation of the line is $x = -3$.



EXERCISE 2.03 ANSWERS ON P. 578

Graphing linear equations UFRC

1 Graph each linear equation on a number plane, and find:

i its x-intercept **ii** its y-intercept.

a $y = 3x - 1$ **b** $y = 2x + 5$ **c** $y = -x + 4$

d $y = -2x - 2$ **e** $y = 4x$ **f** $y = \frac{x}{2} + 3$

EXAMPLE 7

2 Test whether the point $(3, -1)$ lies on each line.

a $y = 2x - 5$ **b** $x - y = 4$ **c** $y + 2x = 5$

d $y = x - 4$ **e** $x + y = 5$ **f** $3x + y + 8 = 0$

EXAMPLE 8

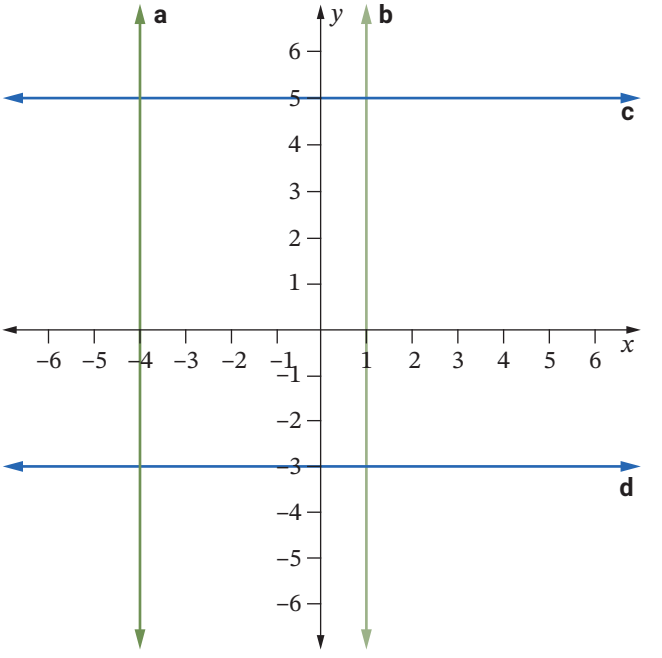
3 Which point lies on the line $y = 6x - 5$? Select the correct answer **A**, **B**, **C** or **D**.

A $(-1, 11)$ **B** $(3, -13)$ **C** $(-2, -17)$ **D** $(-5, 25)$

Foundation Standard Complex

EXAMPLE 9

4 Find the equation of each line shown.



5 Graph each set of lines on a number plane.

a $x = 2\frac{1}{2},\ y = -3,\ y = 1$ **b** $x = 6,\ y = -2,\ x = -\frac{1}{2}$

EXAMPLE 10

6 Find the equation of the line that is: R C

a horizontal and passes through the y-axis at 2

b vertical with an x-intercept of 4

c parallel to the y-axis and passes through the point $(-1, 4)$

d parallel to the x-axis and passes through the point $(0, -2)$

e 3 units above the x-axis

f 1 unit to the left of the y-axis

g drawn through the points $(-1, 6)$ and $(2, 6)$

h drawn through the points $(-1, 8)$ and $(-1, 2)$.

7 Which point lies on the line $4x + y = 1$? Select **A**, **B**, **C** or **D**.

A $(-1, 5)$ **B** $(-2, 7)$ **C** $(6, 9)$ **D** $\left(-\frac{1}{2}, 1\right)$

8 Which line is horizontal and passes through the point $(8, -2)$? Select **A**, **B**, **C** or **D**.

A $y = 8$ **B** $x = 8$ **C** $y = -2$ **D** $x = -2$

9 **a** What is another name for the line $y = 0$? C

b What is another name for the line $x = 0$? C

Foundation Standard Complex

9780170479622

Technology

Graphing $y = mx + c$

1 Use dynamic geometry software to graph the 4 lines $y = 3x + 2$, $y = 5x + 2$, $y = -2x + 2$ and $y = -0.1x + 2$. Make each line a different colour.

2 Find the gradient (slope) and y-intercept of each line and record your results in a table like this.

Equation	Gradient	y-intercept

3 What do you notice about your results?

4 Repeat the steps above for each set of equations.

a $y = -4x$ $y = -4x + 1$ $y = -4x - 10$

b $y = 2x + 3$ $y = -7x + 3$ $y = 0.2x + 3$

c $y = -x - 1$ $y = -x + 2$ $y = -x - 4$

5 For each set of lines drawn in question **4**, complete a table as shown in Step **2**.

6 What do you notice about each set of lines? Identify any key features of each set of graphs.

2.04 The gradient–intercept equation $y = mx + c$

Videos
Graphing $y = mx + c$
Gradient and y-intercept of a line
The gradient-intercept formula

Interactive
Graphing slope-intercept

Puzzles
Equation of a line
Equations in gradient form

Spreadsheet
Drawing straight lines: $y = mx + c$

The gradient–intercept form of a linear equation

The equation of a straight line is $y = mx + c$, where m is the **gradient** and c is the **y-intercept**.

For this reason, $y = mx + c$ is also called the **gradient–intercept form** of a linear equation.

Example 11

Find the gradient and y-intercept of the line with equation:

a $y = -4x + 9$ **b** $y = 10 - 6x$ **c** $y = \frac{5x+4}{2}$ **d** $3x + 2y - 6 = 0$

Solution

a $y = -4x + 9$ is in the form $y = mx + c$.

$\therefore$ Gradient $m = -4$ and y-intercept $c = 9$.

b $y = 10 - 6x$ can be rewritten as $y = -6x + 10$.

$\therefore$ Gradient $m = -6$ and y-intercept $c = 10$.

c $y = \frac{5x+4}{2}$ can be rewritten as $y = \frac{5x}{2} + \frac{4}{2} = \frac{5x}{2} + 2$.

$\therefore$ Gradient $m = \frac{5}{2}$ and y-intercept $c = 2$.

d $3x + 2y - 6 = 0$ can be rearranged in the form $y = mx + c$.

Make y the **subject** of the equation: 'solve' the equation for y.

$$3x + 2y - 6 = 0$$
$$3x + 2y - 6 - 3x = 0 - 3x$$
$$2y - 6 = -3x$$
$$2y - 6 + 6 = -3x + 6$$
$$2y = -3x + 6$$
$$\frac{2y}{2} = \frac{-3x+6}{2}$$
$$y = \frac{-3x}{2} + 3$$

$\therefore$ Gradient $m = -\frac{3}{2}$ and y-intercept $c = 3$.

PATH

Finding the x- and y-intercepts of a line

We can find the x- and y-intercepts of a line from its equation.

Since any point on the x-axis has a y-coordinate of 0, we can substitute $y = 0$ into the equation to find the x-intercept.

Similarly, any point on the y-axis has an x-coordinate of 0, so we can substitute $x = 0$ into the equation to find the y-intercept.

Finding the x- and y-intercepts of a line

- To find the x-intercept, substitute $y = 0$ and solve the equation
- To find the y-intercept, substitute $x = 0$ and solve the equation

Example 12

Find the x- and y-intercepts of each line.

a $y = 3x - 5$ **b** $2x + y - 8 = 0$

Solution

a For x-intercept, substitute $y = 0$:

$0 = 3x - 5$

$5 = 3x$

$x = \frac{5}{3}$

For y-intercept, substitute $x = 0$:

$y = 3(0) - 5$

$y = -5$

Also, y-intercept for $y = 3x - 5$ must be $c = -5$.

b For x-intercept, substitute $y = 0$:

$2x + 0 - 8 = 0$

$2x = 8$

$x = 4$

For y-intercept, substitute $x = 0$:

$2(0) + y - 8 = 0$

$y = 8$

Example 13

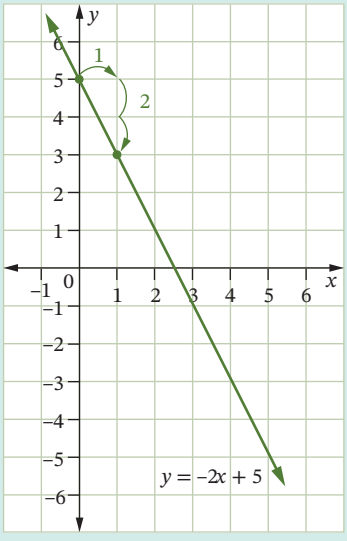
Graph each linear equation by finding the gradient and y-intercept first.

a $y = -2x + 5$ **b** $y = \frac{3}{4}x - 2$

Solution

a $y = -2x + 5$ has a gradient of -2 and a y-intercept of 5.

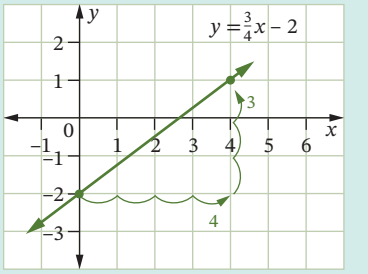
- Plot the y-intercept 5 on the y-axis.
- Make a gradient of $-2 = \frac{-2}{1}$ by moving **across** 1 unit (**run**) and **down** 2 units (**'negative' rise**) and marking the point at (1, 3).
- Rule a line through this point and the y-intercept.



Don't forget to label the line with its equation '$y = -2x + 5$'

b $y=\frac{3}{4}x-2$ has a gradient of $\frac{3}{4}$ and a y-intercept of -2.

- Plot the y-intercept -2 on the y-axis.
- Make a gradient of $\frac{3}{4}$ by moving **across** 4 units (**run**) and **up** 3 units (**rise**) and marking the point at (4, 1).
- Rule a line through this point and the y-intercept.



Example 14

Test whether each line is parallel to $y = -2x + 3$.

a $y = 2x + 3$ **b** $y = -2x + 1$ **c** $y = -2x$ **d** $y = 5x + 3$

Solution

Parallel lines have the **same gradient**. The line $y = -2x + 3$ has the gradient $m = -2$.

a $y = 2x + 3$ has a different gradient, 2, so it is not parallel to $y = -2x + 3$.

b $y = -2x + 1$ has the same gradient, -2, so it is parallel to $y = -2x + 3$.

c $y = -2x$ has the same gradient, -2, so it is parallel to $y = -2x + 3$.

d $y = 5x + 3$ has a different gradient, 5, so it is not parallel to $y = -2x + 3$.

$\therefore$ The lines **b** $y = -2x + 1$ and **c** $y = -2x$ are parallel to $y = -2x + 3$.

EXERCISE 2.04 ANSWERS ON P. 579

The gradient–intercept equation $y = mx + c$ U F R

EXAMPLE 11

1 Find the gradient and y-intercept of each line.

a $y = 3x - 2$ **b** $y = -2x + 7$ **c** $y = x + 4$ **d** $y = 9 - x$

e $y=\frac{3x}{4}+6$ **f** $y = x$ **g** $y=\frac{x}{2}-11$ **h** $y=\frac{2x+18}{3}$

i $y=\frac{-24-x}{3}$ **j** $y = 2(x - 3)$ **k** $11 - 3x = y$ **l** $\frac{2x-7}{2}=y$

PATH

EXAMPLE 12

2 Find the x- and y-intercepts of each line in Question **1**.

3 Find the equation of a line with:

a a gradient of 2 and a y-intercept of 1 **b** a gradient of $\frac{3}{4}$ and a y-intercept of 2

c a gradient of -7 and a y-intercept of 5 **d** a gradient of $-\frac{2}{5}$ and a y-intercept of 3

e $m = -2, c = -3$ **f** $m = -3, c = \frac{1}{2}$

Foundation Standard Complex

 9780170479622

4 Graph each linear equation by finding the gradient and y-intercept first. EXAMPLE 13

a $y = 2x + 1$ **b** $y = 3x - 2$ **c** $y = 2x$ **d** $y = \frac{x}{2} - 1$

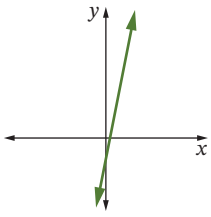
e $y = -2x + 3$ **f** $y = -\frac{3x}{4}$ **g** $y = \frac{-5x+2}{2}$ **h** $y = \frac{3x-20}{5}$

5 Write the equation of a line with a gradient of 2 that passes through the origin. R

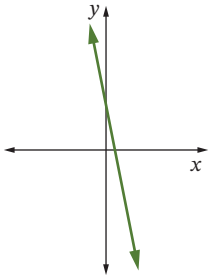
6 Match each equation to its graph below. R

a $y = 4x + 1$ **b** $y = -4x + 1$ **c** $y = -4x - 1$ **d** $y = 4x - 1$

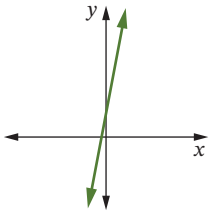
A



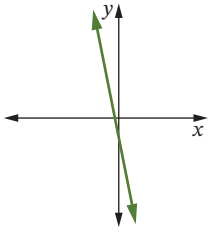
B



C



D



7 For each given line, select the lines that are parallel. There may be more than one answer. R EXAMPLE 14

a $y = x + 6$

A $y = 6x$ **B** $y = 6 - x$ **C** $y = x + 1$ **D** $y = 2x$

b $y = 3x + 10$

A $y = 10x + 3$ **B** $y = 3x - 1$ **C** $y = 1 - 3x$ **D** $y = 4 + 3x$

c $y = \frac{x}{2} + 5$

A $y = 2x - 1$ **B** $y = \frac{x+6}{2}$ **C** $y = 1 - \frac{x}{2}$ **D** $y = x + 2$

d $y = 6$

A $y = 2x + 6$ **B** $y = 6x$ **C** $y = -1$ **D** $y = 10$

e $y = 4x$

A $y = 4x - 2$ **B** $y = 4x + 3$ **C** $y = 4$ **D** $y = 1 - 4x$

f $x = 10$

A $y = 10$ **B** $y = 10x$ **C** $x = 2y$ **D** $x = -6$

8 For each set of linear equations, find a pair of equations whose graphs are parallel lines. R

a $y = 4x + 3$ $y = x + 2$ $y = 4x - 6$ $y = 2x$

b $y = 5x + 1$ $3x - y + 7 = 0$ $y = 3x - 2$ $y = -5x + 2$

Foundation Standard Complex

Quiz
Mental skills 2

Mental skills 2: Maths without calculators ANSWERS ON P. 580

Finding 15%, $2\frac{1}{2}$%, 25% and $12\frac{1}{2}$%

- To find 10% or $\frac{1}{10}$ of a number, divide by 10
- To find 5% of a number, find 10% first, then halve it (since 5% is half of 10%).
- So to find 15% of a number, find 10% and 5% of the number separately, then add the answers together.

1 Study each example.

a $15\% \times 80 = (10\% \times 80) + (5\% \times 80) = 8 + 4 = 12$

b $15\% \times \$170 = (10\% \times \$170) + (5\% \times \$170) = \$17 + \$8.50 = \$25.50$

c $15\% \times 3600 = (10\% \times 3600) + (5\% \times 3600) = 360 + 180 = 540$

d $15\% \times \$28 = (10\% \times \$28) + (5\% \times \$28) = \$2.80 + \$1.40 = \$4.20$

2 Now find 15% of each amount.

a	120	**b**	\$840	**c**	260	**d**	\$202
e	\$50	**f**	72	**g**	\$180	**h**	400
i	\$1600	**j**	\$22	**k**	6000	**l**	\$350

To find $2\frac{1}{2}$% of a number, first find 5%, then halve it.

3 Study each example.

a $2\frac{1}{2}\% \times 600$

$10\% \times 600 = 60$

$5\% \times 600 = \frac{1}{2} \times 60 = 30$

$2\frac{1}{2}\% \times 600 = \frac{1}{2} \times 30 = 15$

b $2\frac{1}{2}\% \times \$820$

$10\% \times \$820 = \$82$

$5\% \times \$820 = \frac{1}{2} \times 82 = \$41$

$2\frac{1}{2}\% \times \$820 = \frac{1}{2} \times \$41 = \$20.50$

4 Now find $2\frac{1}{2}$% of each amount

a	400	**b**	6640	**c**	\$2000	**d**	\$880
e	1500	**f**	\$232	**g**	5400	**h**	\$904

To find 25% of a number, halve the number twice as $25\% = \frac{1}{4}$.

5 Study each example.

a $25\% \times 700$

$50\% \times 700 = \frac{1}{2} \times 700 = 350$

$25\% \times 700 = \frac{1}{2} \times 350 = 175$

b $25\% \times \$86$

$50\% \times \$86 = \frac{1}{2} \times \$86 = \$43$

$\therefore 25\% \times \$86 = \frac{1}{2} \times \$43 = \$21.50$

6 Now find 25% of each amount.

a	2000	**b**	\$80	**c**	18	**d**	\$25
e	\$324	**f**	\$140	**g**	66	**h**	298
i	\$780	**j**	\$1700	**k**	\$126	**l**	1160

To find $12\frac{1}{2}\%$ of a number, find 25% first, then halve it. In other words, halve 3 times because $12\frac{1}{2}\% = \frac{1}{8}$.

7 Study each example.

a $12\frac{1}{2}\% \times 400$

$50\% \times 400 = \frac{1}{2} \times 400 = 200$

$25\% \times 400 = \frac{1}{2} \times 200 = 100$

$12\frac{1}{2}\% \times 400 = \frac{1}{2} \times 100 = 50$

b $12\frac{1}{2}\% \times \$144$

$50\% \times \$144 = \frac{1}{2} \times \$144 = \$72$

$25\% \times \$144 = \frac{1}{2} \times \$72 = \$36$

$12\frac{1}{2}\% \times \$144 = \frac{1}{2} \times \$36 = \$18$

8 Now find $12\frac{1}{2}\%$ of each amount.

a	1280	**b**	\$12	**c**	60	**d**	\$260
e	\$540	**f**	\$250	**g**	304	**h**	1360

The general equation $ax + by + c = 0$ 2.05

A linear equation written in **gradient–intercept form**, such as $y = -\frac{3}{4}x + 2$, can also be written in **general form** ($3x + 4y - 8 = 0$). Note that, for the general form, all of the terms on the left-hand side of the equation are written with no fractions, and only 0 is on the right-hand side. Sometimes the general form is neater and more convenient.

PATH

Worksheet
Parallel and perpendicular lines

The general form of a linear equation

The **general form of a linear equation** is written as $ax + by + c = 0$, where a, b and c are integers and a is positive.

PATH

Video
The general equation $ax + by + c = 0$

Puzzle
Linear equations code puzzle

Example 15

Write each linear equation in general form.

a $y = 6x + 2$ **b** $y = -\frac{2}{3}x + 2$ **c** $y = 2x - \frac{3}{5}$

Solution

a $y = 6x + 2$

$0 = 6x - y + 2$ Subtracting y from both sides

$6x - y + 2 = 0$ Swapping sides so that 0 appears on the RHS

b $y = -\frac{2}{3}x + 2$

$3y = 3\left(-\frac{2}{3}x + 2\right)$ Multiplying both sides by 3 to remove the fraction

$3y = -2x + 6$

$2x + 3y = 6$ Adding $2x$ to both sides

$2x + 3y - 6 = 0$ Subtracting 6 from both sides

c $y = 2x - \frac{3}{5}$

$5y = 5\left(2x - \frac{3}{5}\right)$ Multiplying both sides by 5

$5y = 10x - 3$

$0 = 10x - 5y - 3$ Subtracting $5y$ from both sides

$10x - 5y - 3 = 0$ Swapping sides so that 0 appears on the RHS

Example 16

Find the gradient and y-intercept of the line whose equation is $5x + 2y - 10 = 0$.

Solution

Rewrite $5x + 2y - 10$ in the form $y = mx + c$.

Make y the subject, solve for y.

$5x + 2y - 10 = 0$

$2y - 10 = -5x$ Subtracting $5x$ from both sides

$2y = -5x + 10$ Adding 10 to both sides

$y = \frac{-5x + 10}{2}$ Dividing both sides by 2

$y = \frac{-5x}{2} + 5$

Aim to have y on its own on the LHS of the equation.

$\therefore$ Gradient: $m = -\frac{5}{2}$

y-intercept: $c = 5$

9780170479622

Example 17

Find the x- and y-intercepts of each line.

a $3x - 2y + 12 = 0$ **b** $x + 3y - 7 = 0$

Solution

a For x-intercept, substitute $y = 0$:

$3x - 2(0) + 12 = 0$

$3x = -12$

$x = -4$

For y-intercept, substitute $x = 0$:

$3(0) - 2y + 12 = 0$

$-2y = -12$

$y = 6$

b For x-intercept, substitute $y = 0$:

$x + 3(0) - 7 = 0$

$x = 7$

For y-intercept, substitute $x = 0$:

$0 + 3y - 7 = 0$

$3y = 7$

$y = \frac{7}{3}$

EXERCISE 2.05 ANSWERS ON P. 580

The general equation $ax + by + c = 0$ UFRC

1 Write each linear equation in general form. R C

EXAMPLE 15

a $y = x + 2$ **b** $y = 3x - 1$ **c** $y = 8 + 5x$

d $x + 2y = 3$ **e** $x - 2y = 6$ **f** $y = 8x + 2$

g $y + 3 = 6x$ **h** $2y = x - 6$ **i** $y = \frac{3}{5}x + 2$

2 Find the gradient and y-intercept of the line with equation: R C

EXAMPLE 16

a $2x + y = 6$ **b** $8x - 2y = 10$ **c** $3x - 2y + 4 = 0$

d $y + 2x - 1 = 0$ **e** $2x + y + 5 = 0$ **f** $4x + 3y - 12 = 0$

3 Find the gradient, m, and the y-intercept, c, of the line with equation $x - 3y + 5 = 0$. Select the correct answer **A**, **B**, **C** or **D**.

A $m = -1, c = 5$ **B** $m = \frac{1}{3}, c = \frac{5}{3}$ **C** $m = 1, c = -5$ **D** $m = \frac{1}{3}, c = -\frac{5}{3}$

4 Find the x- and y-intercepts of each line in Question **2**.

EXAMPLE 17

5 Which statement is FALSE about the line whose equation is $3x + y - 6 = 0$? Select **A**, **B**, **C** or **D**. R C

A Its gradient is -3. **B** Its y-intercept is -6.

C Its x-intercept is 2. **D** It is parallel to the line $y = -3x$.

Foundation Standard Complex

2.06 Finding the equation of a line

Video
Finding the equation of a line

Worksheets
Finding the equation of a line

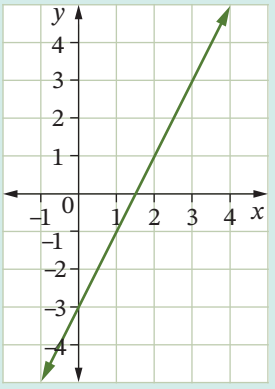
A page of lines

Puzzle
Straight-line equations

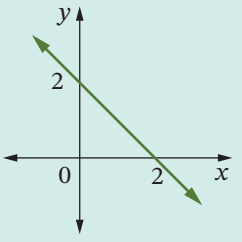
Example 18

Find the equation of each line.

a



b



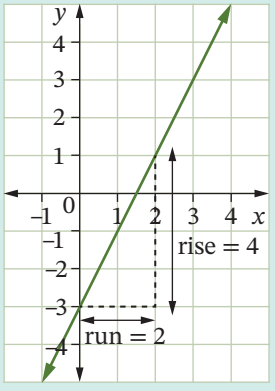
Solution

a Select 2 points on the line to find the gradient, say (0, −3) and (2, 1).

Gradient: $m = \frac{\text{rise}}{\text{run}} = \frac{4}{2} = 2$ from the graph

y-intercept: $c = -3$

∴ The equation of the line is $y = 2x - 3$. $y = mx + c$



We can check that this equation is correct for any point on the line, say (3, 3).

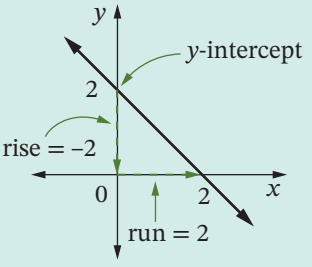
When $x = 3$, $y = 2 \times 3 - 3 = 3$.

b Find the gradient of the line passing through (0, 2) and (2, 0).

Gradient: $m = \frac{\text{rise}}{\text{run}} = \frac{-2}{2} = -1$ from the graph

y-intercept: $c = 2$

∴ The equation of line is $y = -x + 2$. $y = mx + c$

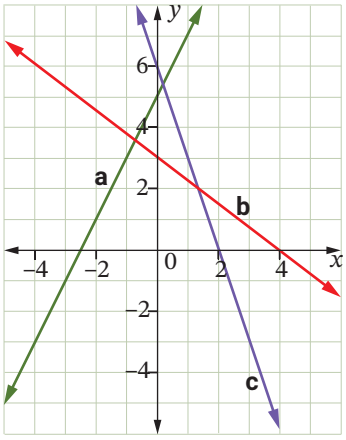


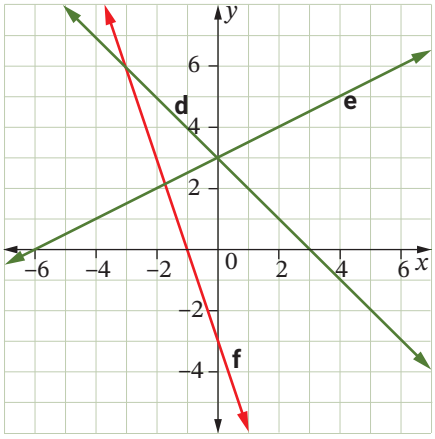
9780170479622

EXERCISE 2.06 ANSWERS ON P. 580

Finding the equation of a line U F R C

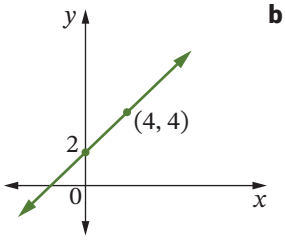
1 Find the equation of each line. R C



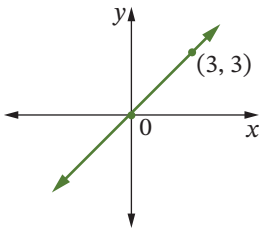


2 Find the equation of each line. R C

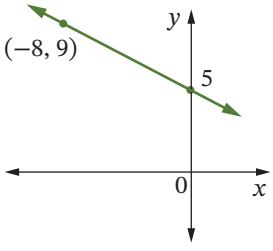
a



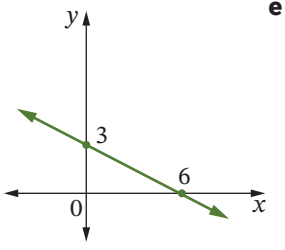
b



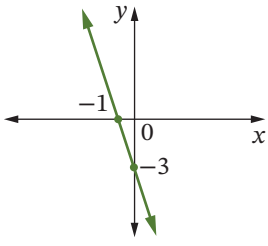
c



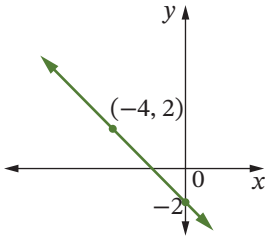
d



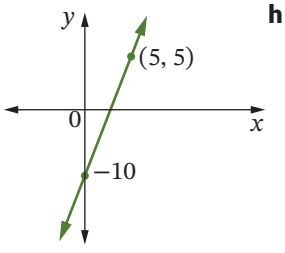
e



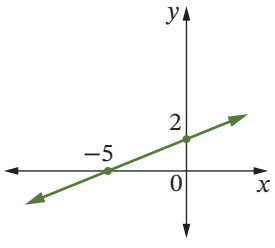
f



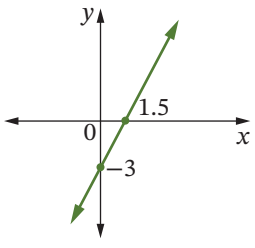
g



h



i

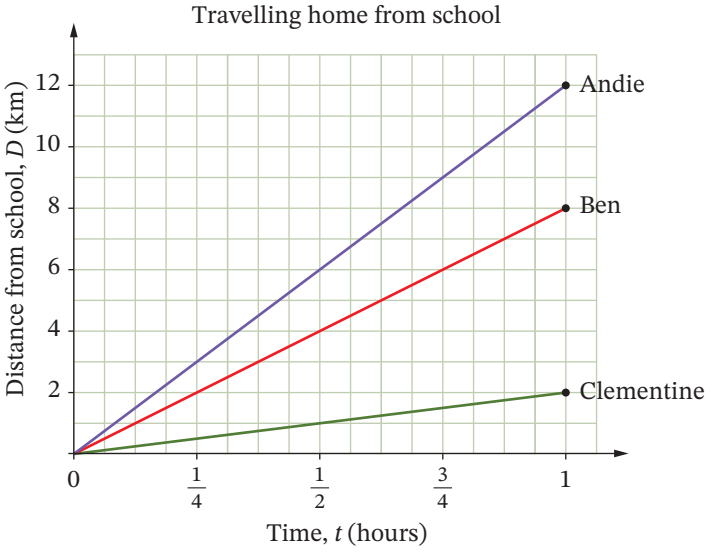


Foundation | Standard | Complex

Investigation

Travelling home

Andie, Ben and Clementine live in a rural area and travel to their nearest school in different ways. This distance-time graph shows their trips home after school.



1. Calculate the gradient of each line.
2. What does the gradient of each line represent?
3. How long did the 3 people take to travel home?
4. What was the speed of each person?
5. They travelled home by walking, jogging or cycling. Which person did what?

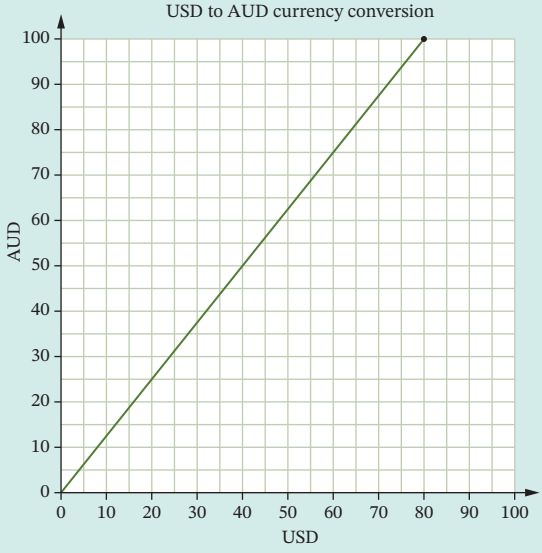
2.07 Applying linear equations

Video
Applying linear functions

Linear equations can be used to model real-life situations such as the cost of a taxi trip or the exchange rate of currencies between 2 countries or the amount of water in a swimming pool when it is being drained. Instead of using x and y as our variables in the linear equation $y = mx + c$, we can use other letters of the alphabet to represent quantities.

Example 19

This conversion graph shows the exchange rate between USD (US dollars) and AUD (Australian dollars)



a Find the gradient and vertical intercept of the line.

b What does the gradient represent?

c Write the equation of the line in the form $A = mU + c$ using U for USD and A for AUD.

d Use the graph to convert:

i 60 USD to AUD

ii 25 AUD to USD

Solution

a Line goes from (0, 0) to (80, 100).

Gradient $= \frac{100-0}{80-0} = 1.25$

Vertical intercept $= 0$ (from graph)

Vertical intercept is the general name for the y-intercept when a variable other than y is used.

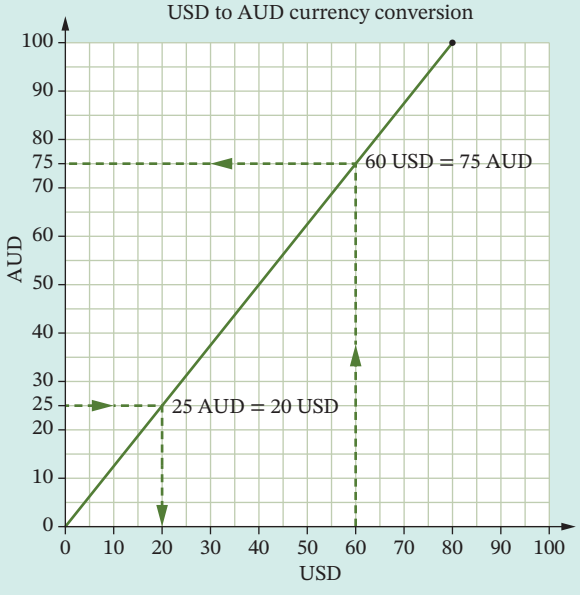
b The gradient, 1.25, represents the exchange rate between USD and AUD (1 USD = 1.25 AUD)

c $A = mU + c$ and we know that $m = 1.25$ and $c = 0$.

We're replacing x with U because U is on the horizontal (x) axis and y with A because A is on the vertical (y) axis.

The equation of the line is $A = 1.25U + 0$, which simplifies to $A = 1.25U$.

d Using the graph:



i 60 USD = 75 AUD

ii 25 AUD = 20 USD

Applying linear equations

- The **gradient** represents the **rate of increase or decrease** in the situation.
- For an increase, the gradient is positive and for a decrease, the gradient is negative.
- The *y*-**intercept** or **vertical intercept** represents the **starting value** in the situation.

EXERCISE 2.07 ANSWERS ON P. 580

Applying linear equations U F R C

The questions in this exercise may also be completed with the aid of a spreadsheet and graphing software.

1 Write an equation in the form $y = mx + c$ using the given change of variables and values. C

a $y \to D, x \to h, m = 30, c = 5$

b $y \to R, x \to t, m = 0.5, c = -8$

c $y \to P, x \to n, m = -50, c = 0$

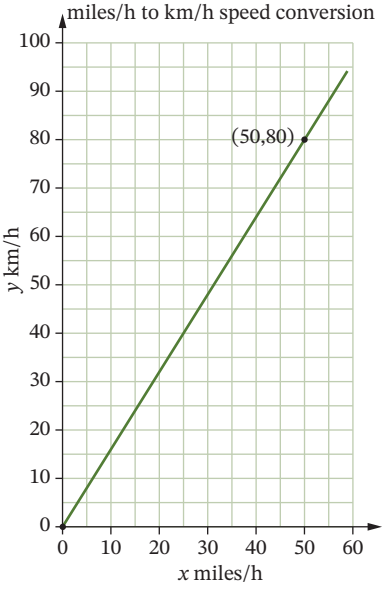
d $y \to F, x \to C, m = 1.8, c = 32$

e $y \to H, x \to w, m = 0.8, c = 0$

f $y \to V, x \to T, m = -40, c = 200$

EXAMPLE 19

2 This conversion graph converts between speeds in miles per hour to km/h. R C



a Use the graph to find what speed in km/h is equivalent to 25 miles/h.

b Find the gradient and vertical intercept of the line.

c What does the gradient represent?

d Find the equation of the line.

3 The distance d km travelled by a car moving at a constant speed of 80 km/h after t hours is given by the equation $d = 80t$. R C

a Copy and complete this table of values using the equation, then graph the equation on a number plane.

Time, *t* hours	0	1	2	3	4
Distance, *d* km					

b Find the gradient of the line.

c What does the gradient represent?

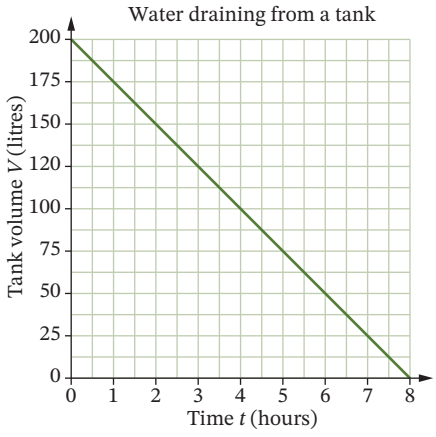
d Use the graph to find:

i the distance travelled by the car after $5\frac{1}{2}$ hours

ii the time taken for the car to travel 40 km

Foundation Standard Complex

4 Water is draining out of a tank. The volume of water in the tank is shown by this graph. R C



- **a** What is the gradient and vertical intercept of the line?
- **b** What does the vertical intercept represent?
- **c** What does the gradient represent?
- **d** Why is the gradient negative?
- **e** Find the equation of the line using the variables V and t.

5 Vinay is an emergency electrician charges a $100 callout fee plus $75 per hour thereafter. This can be represented by the equation $C = 75t + 100$ where C represents the cost in dollars and t represents the number of hours. R C

- **a** Draw a graph to represent this equation.
- **b** Find the gradient and vertical intercept of the line from the graph and the equation.
- **c** Use the graph or equation to find:
  - **i** the charge for a 4-hour job
  - **ii** the time taken if the charge was $700

6 Greta bought a new car for $28 000. Its value depreciates at a constant rate of $4000 per year. R C

- **a** Copy and complete this table, then graph the values.

Number of years, n	0	1	2	3
Value, $V	28 000	24 000		

- **b** Extend your graph to find after how many years the car will be worth nothing.
- **c** Find the gradient and vertical intercept of the line.
- **d** What do the gradient and vertical intercept of the line represent?
- **e** Find the equation of the line using V and n.

7 Greta's car has a fuel consumption of 8 L/100 km. This means the car uses 8 litres of fuel for every 100 km travelled.

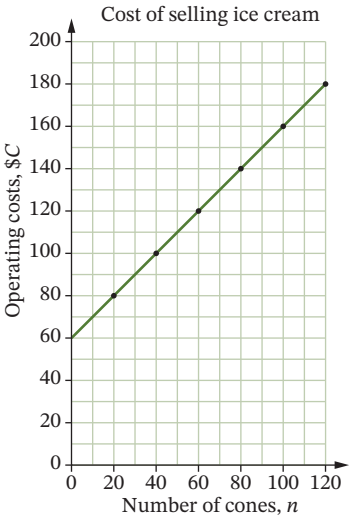
- **a** Copy and complete this table, then graph the values. R C

Distance travelled, d km	100	200	300	400	500
Petrol used, P L					

- **b** Find the gradient and vertical intercept of the line.
- **c** What does the gradient represent?
- **d** Find the equation of the line.
- **e** Use the graph to find the amount of petrol used to travel 350 km.
- **f** Verify your answer in e using the equation of the line.

Foundation Standard Complex

8 Year 10 sells ice cream at lunchtime as a fund raiser. This graph shows how much it *costs* to set up a stall and produce the ice cream cones. R C



a Before any ice cream is sold, how much does it cost to set up the stall? What costs would this represent?

b By calculating the gradient of the line, determine how much it costs to make each ice cream cone.

c Find the equation of the line.

d Calculate the total cost of making 100 ice cream cones, including the set-up costs.

e If each cone is sold for $5, how much profit would be made from selling 100 ice cream cones?

9 Pavla is saving for her end-of-school holiday. She has her birthday money of $500 and plans to add $50 to that amount each week from her part-time job for the next 2 years. R C

a Copy and complete this table of her savings, then graph these values.

Number of weeks (w)	0	10	20	30	40	50
Savings ($S)						

b What is the vertical intercept and what does it represent?

c What is the gradient of the line and what does it represent?

d Write the equation of the line.

e Use the equation to calculate how much Pavla will save after:

i 27 weeks? **ii** 60 weeks? **iii** 2 years?

d If her goal is $4500, how whole weeks will it take to achieve this amount?

10 The formula $F = 1.8C + 32$ is used to convert temperatures in degrees Celsius (°C) to degrees Fahrenheit (°F). R C

a Copy and complete this table and draw a conversion graph.

Celsius, C	−15	−10	0	10	15
Fahrenheit, F					

b Find the gradient and vertical intercept of the line using the graph and formula.

c Use the formula to convert:

i 6°C to °F **ii** 86°F to °C

Verify your answers using the graph.

Foundation Standard Complex

9780170479622

11 Kembla borrowed $1200 from his uncle to buy a kayak. He agrees to pay back $25 a week. R C

a Write an equation to show how much money (A) Kembla still *owes* after n weeks.

b Use the equation to find the amount owing after:

i 35 weeks **ii** 45 weeks

c Graph the equation.

d Why is the gradient of the line negative?

e Use the graph to verify the vertical intercept and the gradient.

f After how many weeks will the loan be repaid?

Investigation

Sausage sizzle

A local football club is organising a sausage sizzle on Saturday to raise money to buy new equipment. It costs $50 to hire a gas bottle to run the barbecue and each sandwich costs $1 to make. They hope to sell 100 sandwiches.



Alamy Stock Photo/Boaz Rottem

1 Copy and complete this table to show the cost of making sausage sandwiches. Include the cost of hiring the gas bottle.

No. of sandwiches (x)	0	10	20	30	40	50	60	70	80	90	100
Cost ($y)	50	60									

2 Find the linear equation (formula) for y that represents the cost of making x sausage sandwiches.

3 Use an appropriate scale to construct a graph that shows the cost of making from $x = 0$ to 100 sandwiches. Label your axes and give your graph an appropriate title.

4 How much will it cost to make 35 sausage sandwiches?

5 How many sandwiches can be made for $132?

6 How much would it cost to make 120 sausage sandwiches?

7 **a** If the club sold all 100 sausage sandwiches for $5 each, how much money would they take?

b How much profit would the club make?

Foundation Standard Complex

2.08 Equations of parallel and perpendicular lines

Worksheet
Writing equations of lines

Puzzles
Linear equations match-up

Equations of parallel lines

Video
Equations of parallel lines

Gradients of parallel and perpendicular lines

If 2 lines with gradients m_1 and m_2 are parallel, then $m_1 = m_2$.

If 2 lines with gradients m_1 and m_2 are perpendicular, then $m_1 \times m_2 = -1$ or $m_2 = -\frac{1}{m_1}$.

Example 20

Find the equation of the line parallel to $y = 8 - 3x$ that passes through the point $(-1, 6)$.

Solution

For $y = 8 - 3x$ (or $y = -3x + 8$), the gradient is $m = -3$.

A line parallel to $y = 8 - 3x$, will also have $m = -3$.

The equation of this line is $y = mx + c = -3x + c$, where c is a constant.

To find the value of c, substitute the point $(-1, 6)$ into the equation:

$y = -3x + c$

$6 = -3 \times (-1) + c$ $\qquad x = -1, y = 6$

$6 = 3 + c$

$c = 3$

$\therefore$ The equation is $y = -3x + 3$.



ngoc tran/Shutterstock.com

9780170479622

Example 21

Find the equation of the line perpendicular to $3x - 4y + 6 = 0$ that passes through the point $(5, 4)$.

Solution

To find the gradient of $3x - 4y + 6 = 0$, first convert it to the form $y = mx + c$:

$3x - 4y + 6 = 0$

$3x + 6 = 4y$

$4y = 3x + 6$

$y = \frac{3x+6}{4}$

$y = \frac{3}{4}x + \frac{3}{2}$ $\quad y = mx + c$

$\therefore$ Gradient $= \frac{3}{4}$

$\therefore$ Gradient of perpendicular line $= \frac{-1}{\left(\frac{3}{4}\right)}$ $\quad$ The negative reciprocal of $\frac{3}{4}$

$= -\frac{4}{3}$

$\therefore$ The equation of this line is $y = -\frac{4x}{3} + c$

To find the value of c, substitute the point $(5, 4)$ into the equation.

$4 = \left(-\frac{4}{3}\right) \times 5 + c$ $\quad x = 5, y = 4$

$= -\frac{20}{3} + c$

$4 + \frac{20}{3} = c$

$c = \frac{32}{3}$

$\therefore$ The equation is $y = -\frac{4x}{3} + \frac{32}{3}$ or $y = \frac{-4x+32}{3}$ or,

converting to the neater general form:

$3y = -4x + 32$

$4x + 3y - 32 = 0$

Video
Equations of perpendicular lines

2.08

EXERCISE 2.08 ANSWERS ON P. 582

Equations of parallel and perpendicular lines U F R C

EXAMPLE 20

1 Find the equation of the line that is parallel to: R C

a $y = 2x + 9$ and has a y-intercept of 4

b $y = 3x$ and has an x-intercept of -2

c $y = 5 - \frac{x}{2}$ and passes through $(-1, 6)$.

d $2x - y = 6$ and passes through $(5, -2)$

e $y = -5x - 8$ and passes through the midpoint of $(3, -10)$ and $(-5, -6)$

f $2y = x - 3$ and passes through $(6, -7)$

EXAMPLE 21

2 Find the equation of a line that is perpendicular to: R C

a $y = \frac{x}{2}$ and has a y-intercept of -2

b $y = -5x$ and has an x-intercept of 1

c $y = 3x - 1$ and passes through the x-axis at 4

d $y = \frac{x-6}{3}$ and passes through $(1, -6)$

e $x + y - 6 = 0$ and passes through $(-4, 2)$

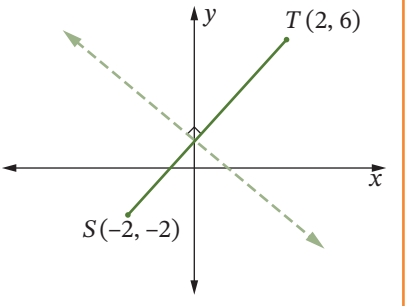
f $3x - y - 9 = 0$ and passes through $(-10, -7)$

3 **a** Find the gradient of interval ST in the diagram.

b Find the midpoint of ST.

c The dotted line is perpendicular to ST and passes through its midpoint. What is its gradient?

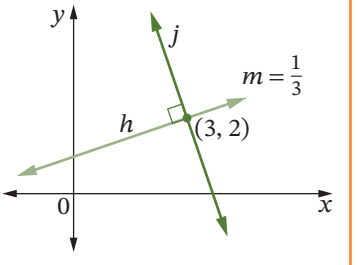
d Find the equation of the dotted line, in the form $y = mx + c$.



4 **a** Find the equation of line h in the diagram.

b Find the gradient of line j (which is perpendicular to line h).

c Find the equation of line j. R C



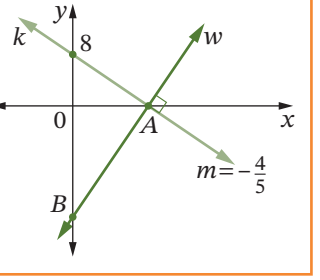
5 **a** Find the equation of line k.

b Find the coordinates of point A.

c Find the gradient of line w.

d Find the equation of line w.

e Find the coordinates of point B. R C



Foundation Standard Complex

9780170479622

Did you know?

Constants

Expressions like 'k is a constant' are often used in mathematics, but constants are also used in areas such as physics, chemistry, biology and astronomy. A constant may be:

- a numerical part of an algebraic expression. For example, in the expression $3x^2 + 5$, the 3 and 5 are constants and 5 is usually called the constant term.
- a quantity that has a fixed value for an expression or calculation. For example, in the equation of a line, $ax + by + c = 0$, the a, b and c are constants (while x and y are variables).
- a number or quantity that does not change in any circumstances. Examples are c (the speed of light) in the formula $E = mc^2$, and π.

Other constants that do not change include Faraday's constant, Planck's constant, Boltzmann's constant, Avogadro's number, 1 astronomical unit, and the gravitational constant.

1 **Find the symbol and value of each of the constants listed above.**

2 **Explain the meaning of the word 'constants' in this statement: 'There are only 2 constants in life—death and taxes'.**

Power plus ANSWERS ON P. 582

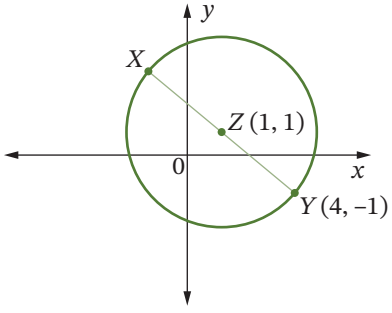
1 A line is drawn through the points $A(0, -2)$ and $B(3, 0)$. The x-coordinate of a point C on AB is 9. Find:

a the gradient of AB **b** the equation of AB **c** the y-coordinate of C.

2 The point $(-1, 6)$ lies on the line $kx + 3y - 13 = 0$, where k is a constant number. Find k.

3 $Z(-1, 3)$ is the midpoint of the interval joining $A(-4, 7)$ and B. Find the coordinates of B.

4 The circle has XY as a diameter and centre Z. What are the coordinates of X?



5 Show that the points (4, 2), (10, –4) and (1, 5) are collinear.

CHAPTER 2 REVIEW

Quiz
Language of maths 2

Puzzle
Graphing lines crossword

Language of maths

axes	distance	exact answer	general form
gradient	gradient–intercept form	horizontal	interval
length	linear	linear equation	midpoint
parallel	perpendicular	reciprocal	rise
run	surd	vertical	vertical intercept
x-axis	x-intercept	y-axis	y-intercept

1 What is the difference between the y-axis and the y-intercept?

2 When finding the length of an interval on a number plane, what is meant by an **exact** answer?

3 What measurement is the fraction given by the vertical rise of a line divided by the horizontal run?

4 Research the everyday meaning of the word **intercept**. Look it up in a dictionary.

5 Describe the property of the gradients of perpendicular lines.

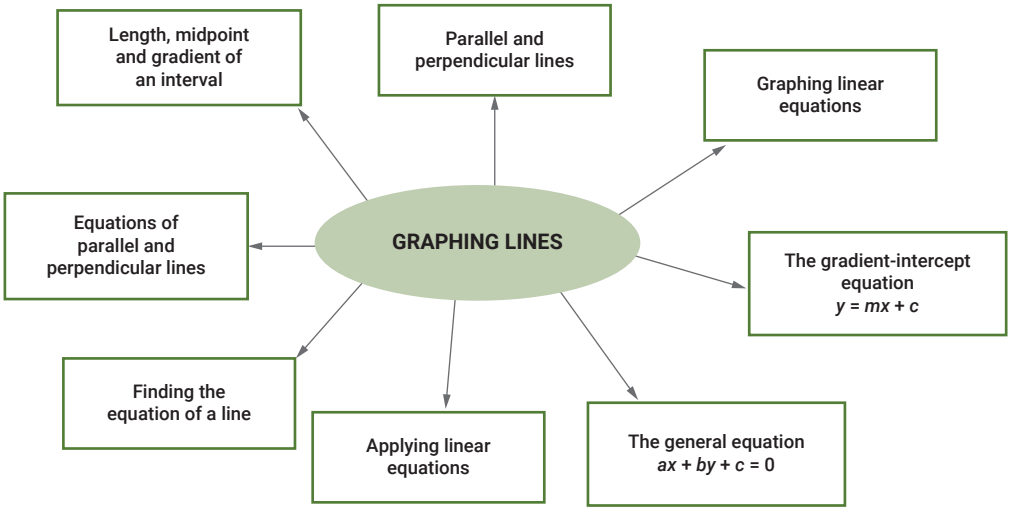
6 What form of the linear equation is $ax + by + c = 0$?

Worksheet
Mind map: Graphing lines

Topic summary

- How can you find the gradient of a line?
- What is $y = mx + c$?
- How can you test whether a pair of lines are parallel?
- What parts of this topic did you find difficult?

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.



 9780170479622

TEST YOURSELF 2

ANSWERS ON P. 582

1 An interval is formed by joining the points $K(5, 6)$ and $L(-7, 2)$. 2.01

a Find, correct to one decimal place, the length of interval KL.

b Find the midpoint of KL.

c Find the gradient of KL.

2 The vertices of a quadrilateral $HJKL$ are $H(-8, -5)$ $J(-1, -2)$ $K(2, 5)$ $L(-5, 2)$. 2.01

a Find the exact length of the sides of the quadrilateral.

b Find the gradient of each side of $HJKL$.

c Find the exact length of the diagonals HK and JL.

d What type of quadrilateral is $HJKL$?

3 A line passes through the points $V(8, -1)$ and $W(10, -2)$. What is the gradient of a line: 2.02

a parallel to VW? **b** perpendicular to VW?

4 Graph the linear equation $y = -5x - 1$ on a number plane. 2.03

5 Which point lies on the line of $3x + y = 2$? Select the correct answer **A**, **B**, **C** or **D**. 2.03

A $(1, 0)$ **B** $(2, 4)$ **C** $(-1, 5)$ **D** $(-1, -5)$

6 What is the equation of the line through $(-2, 3)$ and parallel to the x-axis? Select **A**, **B**, **C** or **D**. 2.03

A $x = -2$ **B** $x = 3$ **C** $y = -2$ **D** $y = 3$

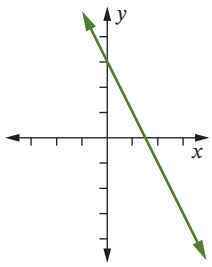
7 Write the gradient, m, and y-intercept, c, for each linear equation. 2.04

a $y = 2x - 10$ **b** $y = 4x + 3$ **c** $y = \frac{4 - 3x}{8}$

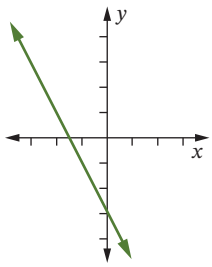
8 Match each equation to its graph below.

a $y = 2x - 3$ **b** $y = -2x - 3$ **c** $y = -2x + 3$ **d** $y = 2x + 3$

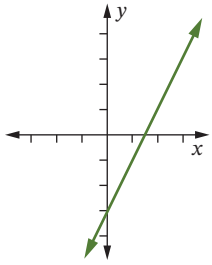
A



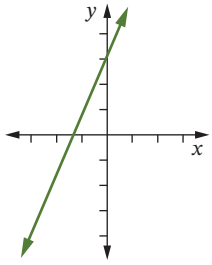
B



C



D



Foundation | Standard | Complex

PATH

9 Find the x- and y-intercepts of each line in Question **7**. 2.04

10 Convert each equation to general form $ax + by + c = 0$. 2.05

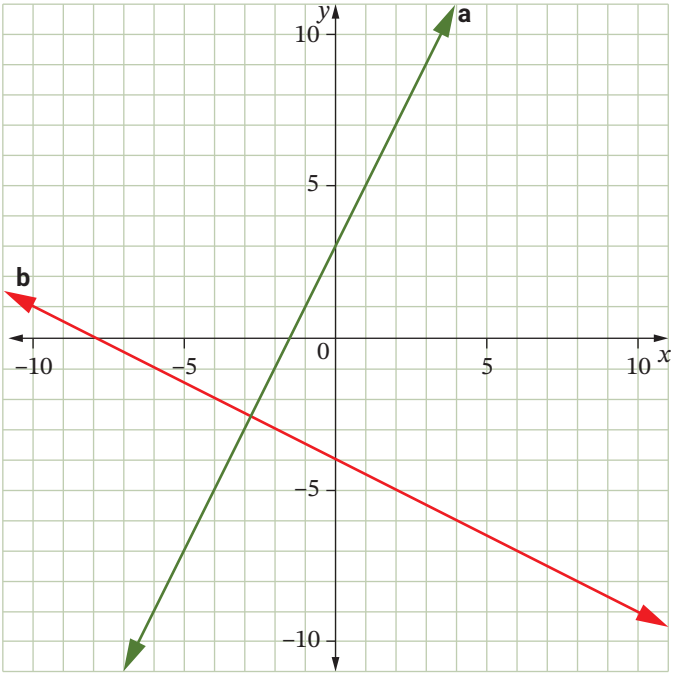
a $y = 3x + 5$ **b** $y = \frac{2x}{5} - 10$ **c** $x = 3y + 6$

11 Rewrite each equation in the form $y = mx + c$, then state the gradient, m, and the y-intercept, c. 2.05

a $x - y + 2 = 0$ **b** $2x - 8y + 8 = 0$ **c** $3x + y - 9 = 0$

12 Find the x- and y-intercepts of each line in Question **11**. 2.05

13 Find the equation of each line and show that they are perpendicular. 2.06



Foundation Standard Complex

9780170479622

14 Seth's Self Storage charges an upfront fee of $50 plus $75 a month thereafter to store items. 2.07

a Copy and complete this table of values, then graph the values.

Number of months, n	0	6	12	18	24
Cost, $\$C$					

b Calculate the gradient of the line.

c What is the vertical intercept of the line?

d Find the equation of the line.

e Use the equation or the graph to find:

i the cost of storing items for 5 months

ii the number of months that items can be stored if the cost is $650

15 Find the equation of a line that is: 2.08

a parallel to $y = 3x + 1$ and passes through the x-axis at 2

b perpendicular to $y = \frac{x}{2}$ and passes through the origin.